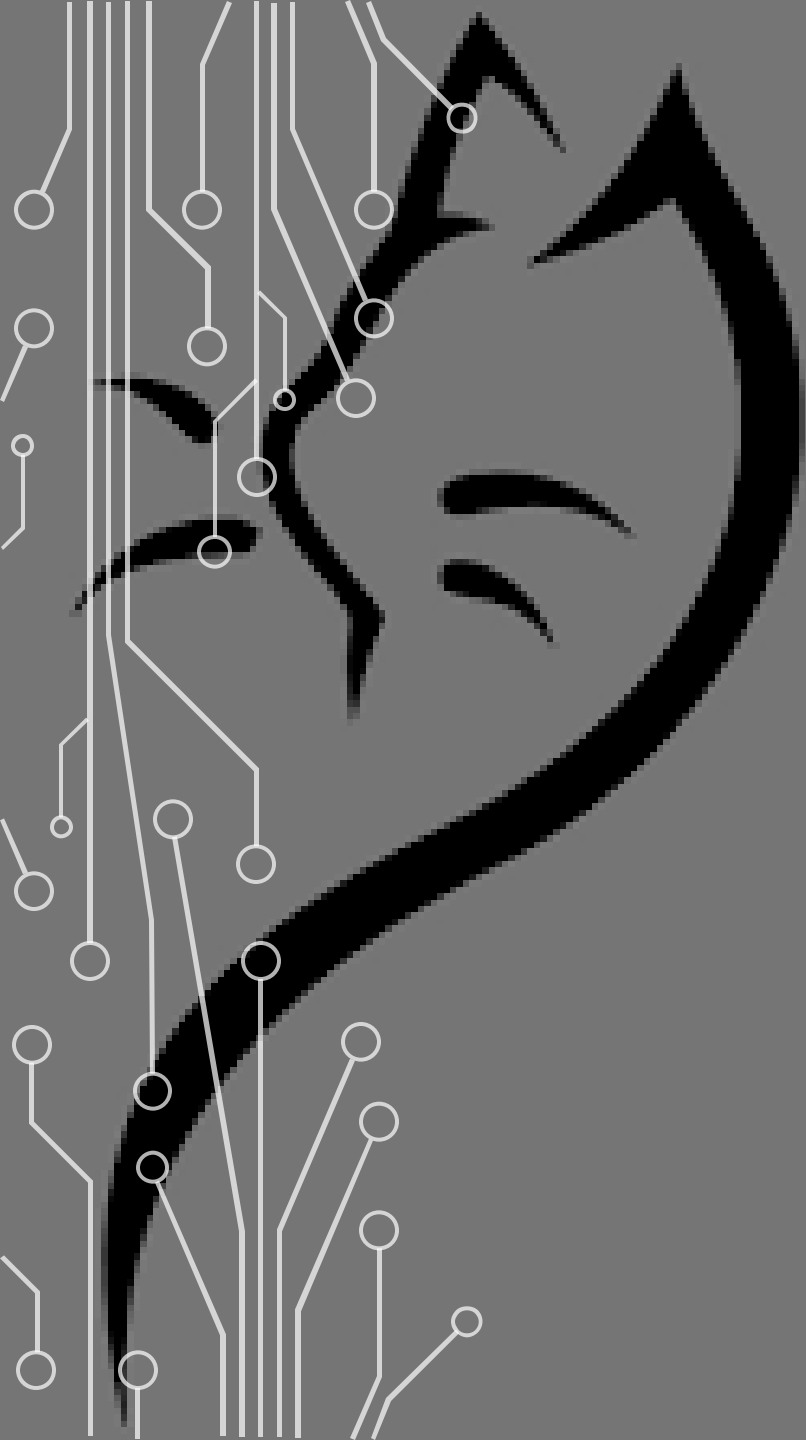




COPYCAT

AI-POWERED SECURE CODE ANALYSIS TOOL (CAT)

Amanda Bradburn — Backend / Security / Integration Developer

Will Davis — AI / LLM Integration Developer

Nolan Biggers — Frontend Developer

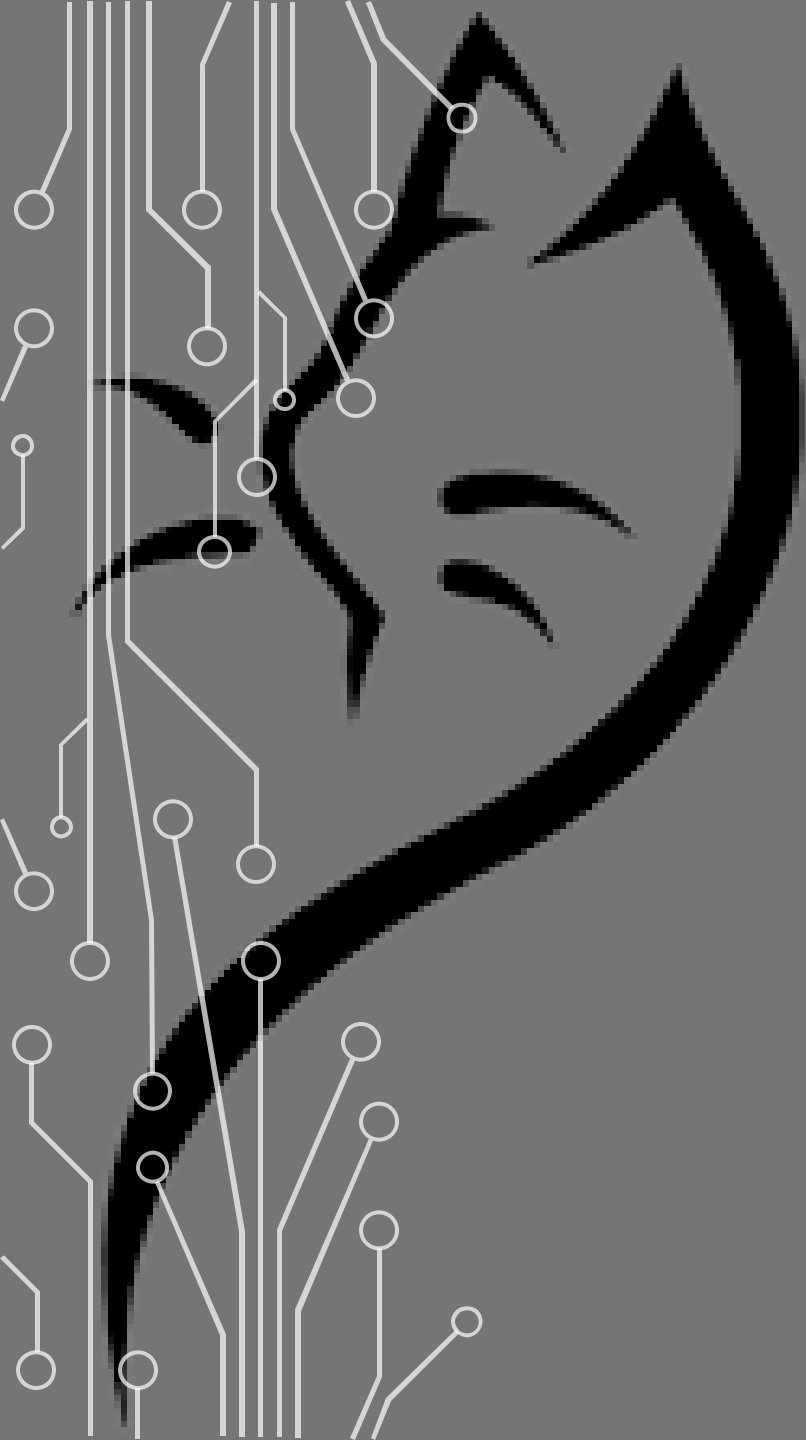
Technologies

- Flask
- Python
- Bandit
- Gemini AI
- GitHub Actions CI/CD

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PROJECT OVERVIEW

PROBLEM

Many beginner developers struggle to identify and understand vulnerabilities in their code.

SOLUTION

CopyCat combines static analysis with AI-generated explanations to help users:

- Detect vulnerabilities
- Understand security risks
- Learn safer coding practices

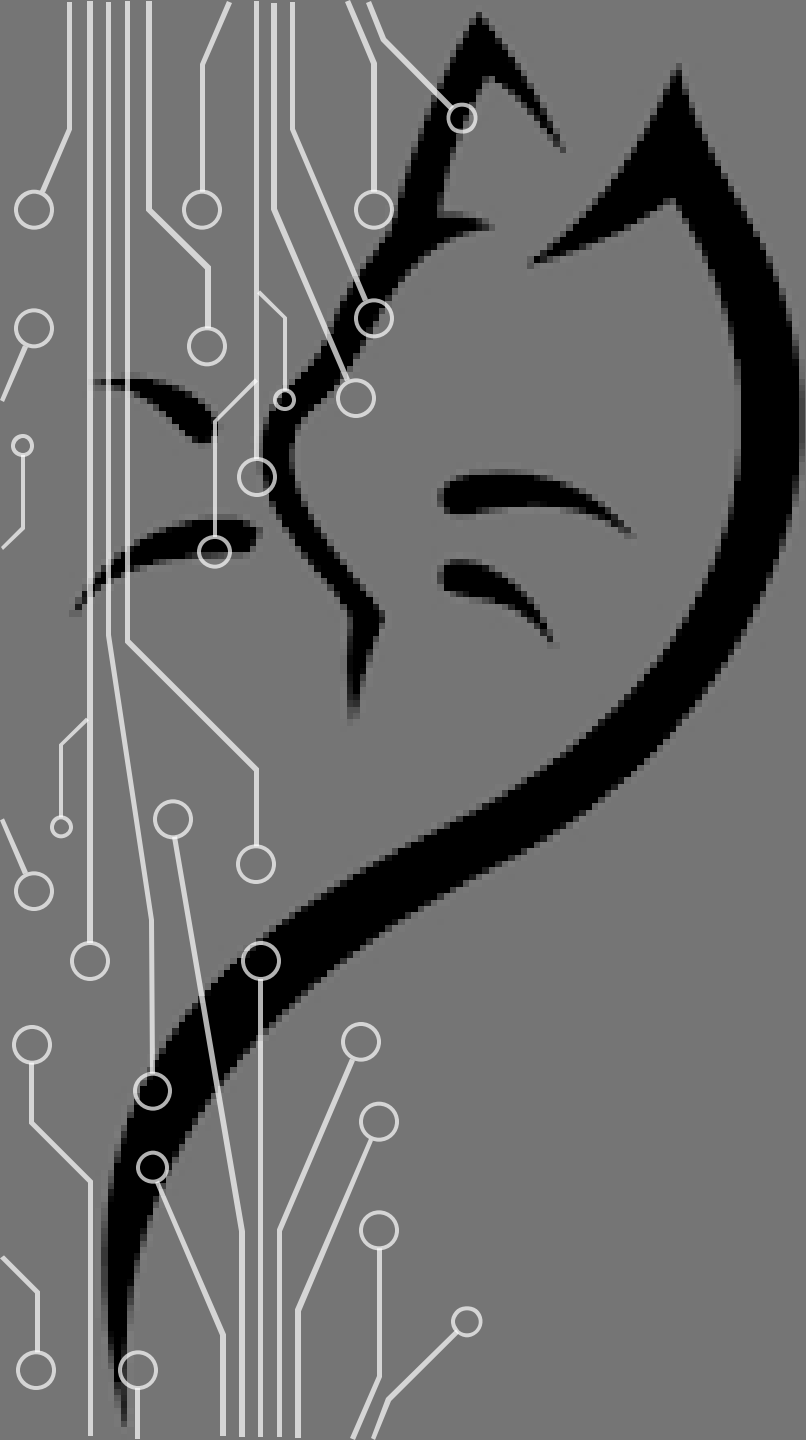
GOAL

Create an easy-to-use educational code analysis tool.

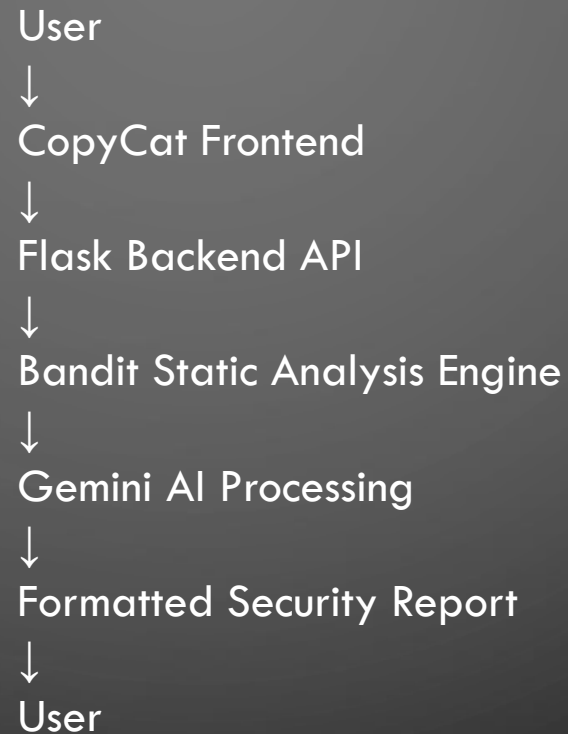
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SYSTEM ARCHITECTURE



FRONTEND

- HTML/CSS/JavaScript UI
- Code submission form
- Displays formatted findings

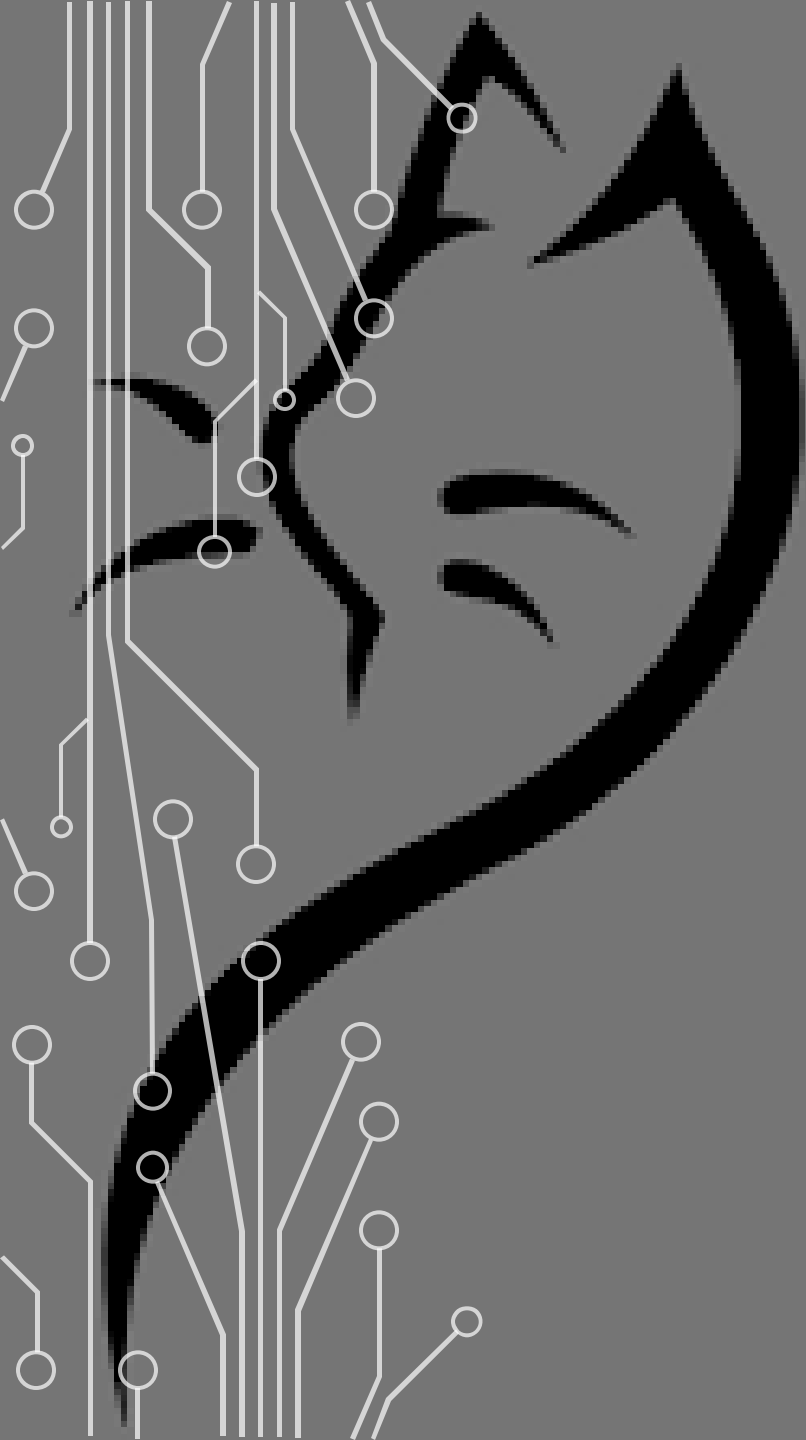
BACKEND

- Flask API
- Handles requests and validation
- Processes Bandit scan results
- Logs backend errors and API failures

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DEVELOPMENT PROCESS

Agile Development

- Incremental milestone-based development
- Continuous testing and debugging
- Frequent feature improvements and fixes

AI Model Configuration

- Default mode uses a lightweight Gemini model for lower latency and reduced cost
- Enhanced AI mode can be enabled for deeper analysis during demonstrations
- API keys and model access are protected using .env

Development Tools

- GitHub version control
- GitHub Actions CI/CD

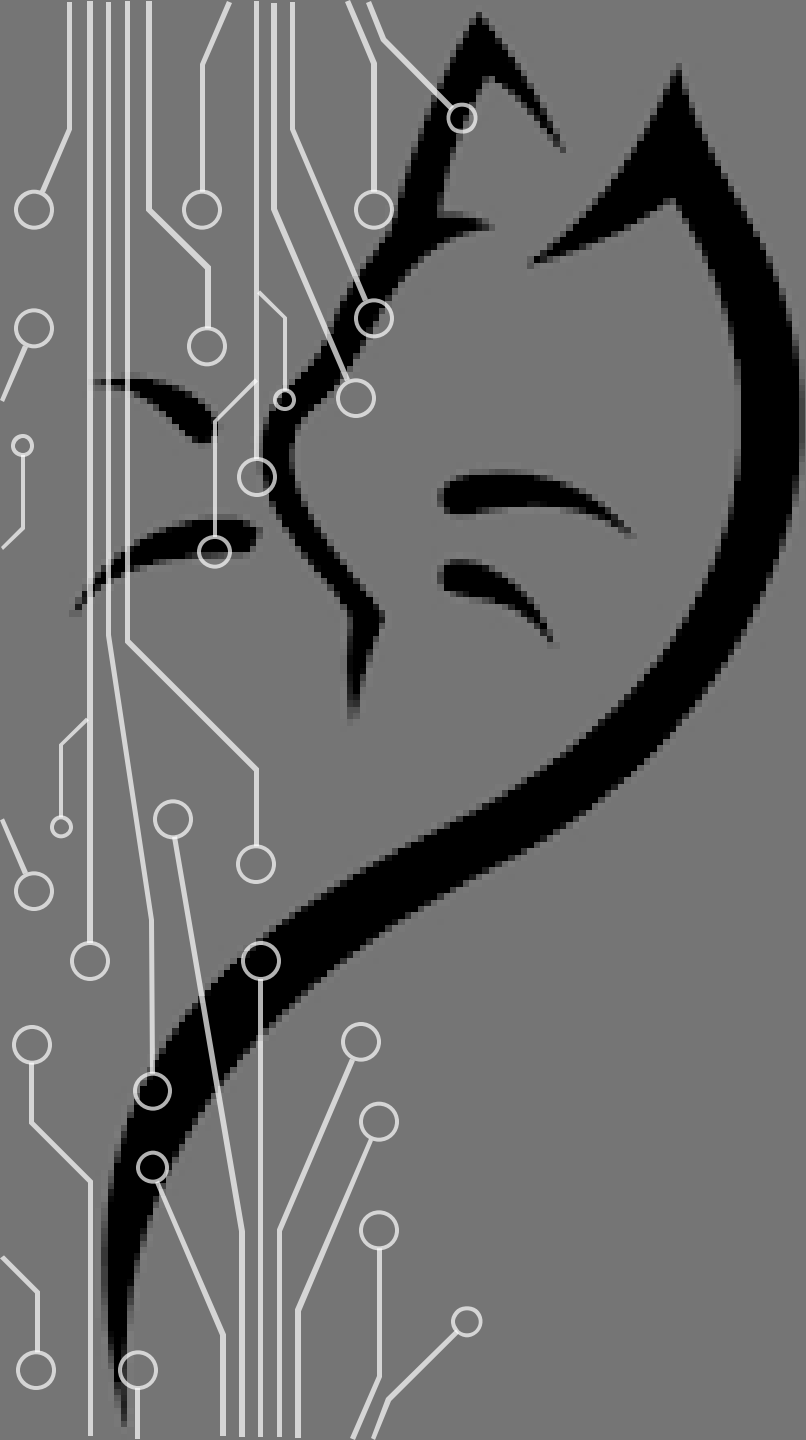
Security Focus

- Security-focused backend hardening

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THREAT MODELING

SECURITY METHODOLOGY

- Static analysis performed using Bandit
- AI explanations grounded using Bandit findings
- Input validation added to reduce malformed requests
- Logging implemented for backend error tracking
- CI/CD used to identify testing and dependency issues

CONCERNS TO DISCUSS

- AI hallucinations
- Unsafe code submissions
- Backend exposure risks
- Need for secure defaults

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MITIGATIONS

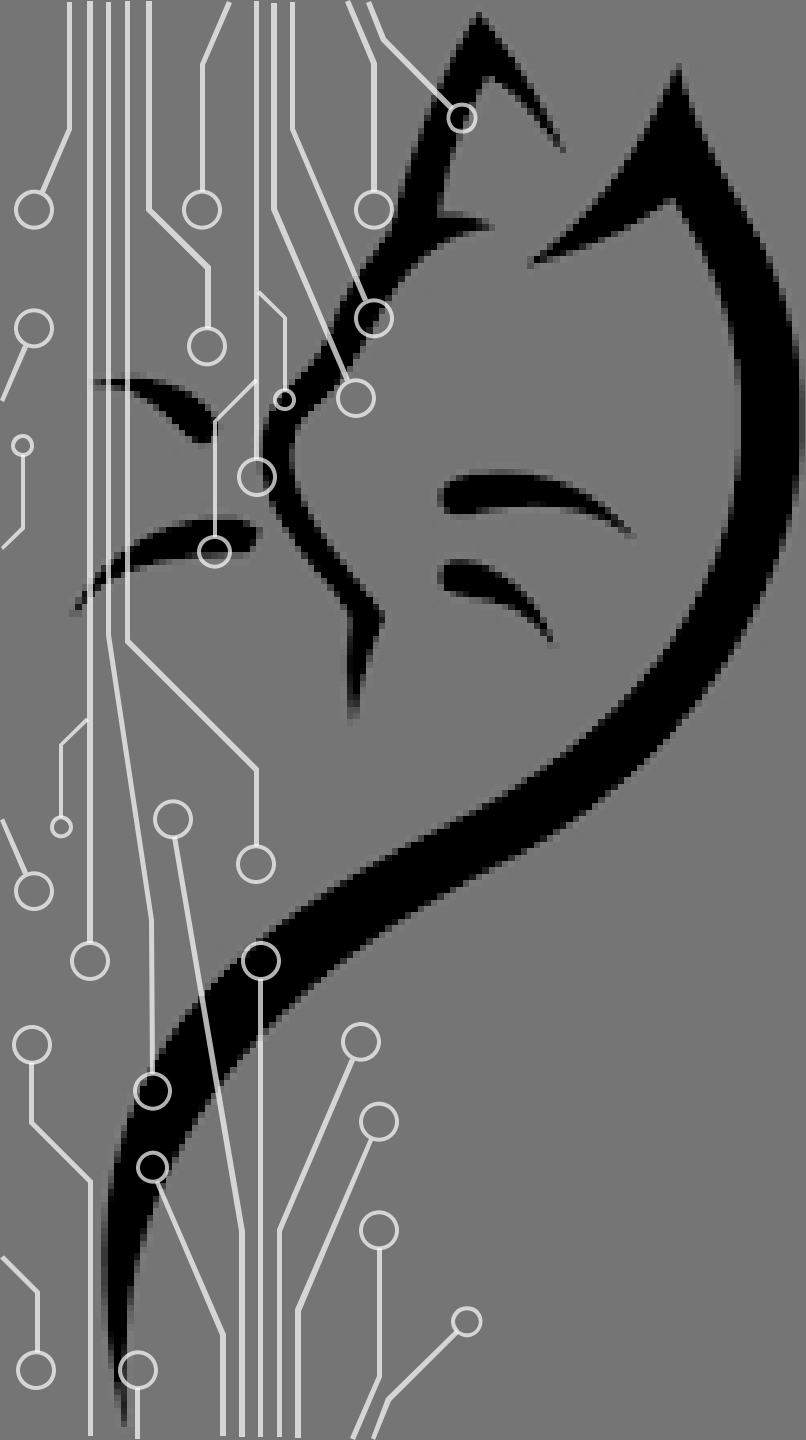
Challenges Encountered

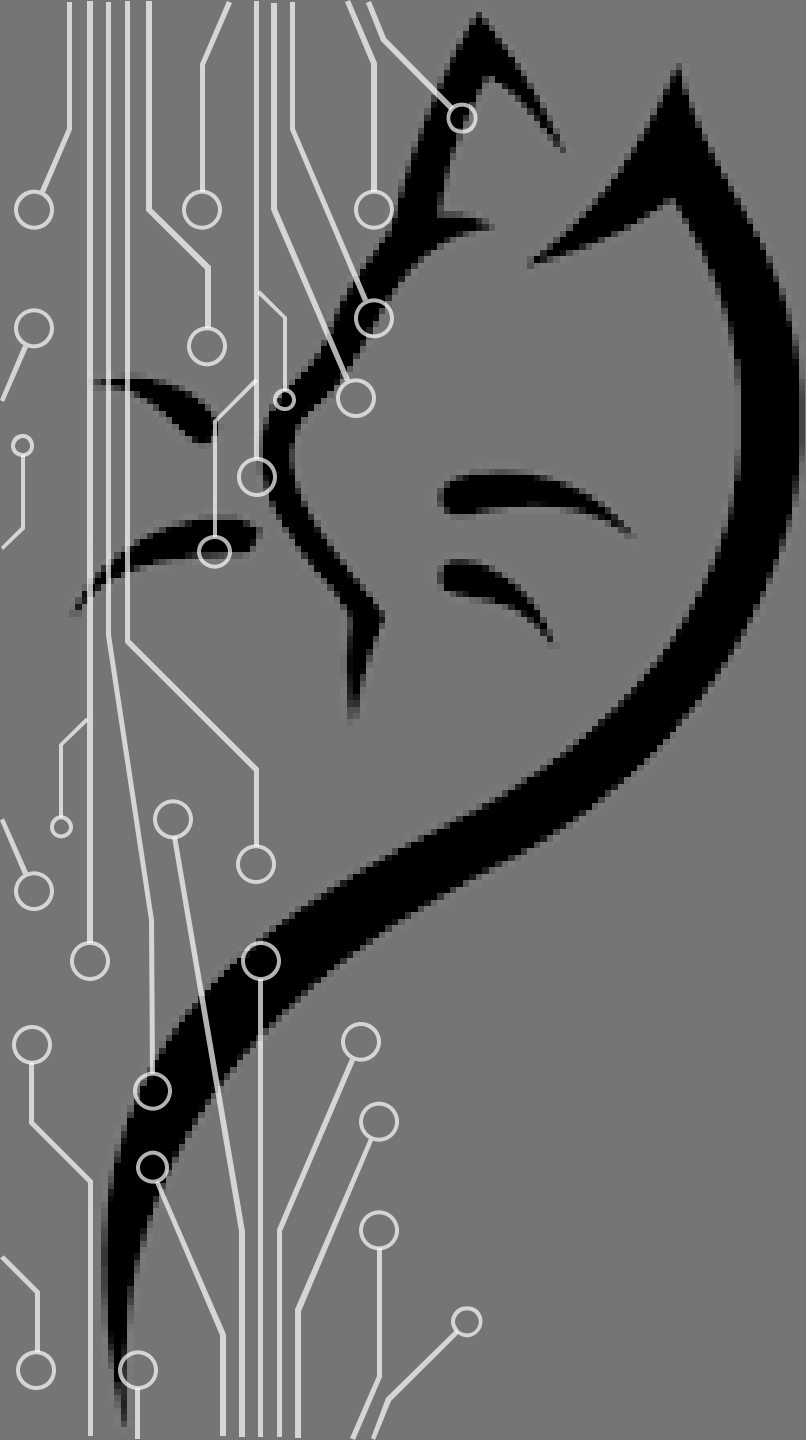
- Bandit grounding before AI explanations
- Input validation added
- Backend logging implemented
- Error handling improvements
- CI/CD testing pipeline
- Limited language support for beta languages

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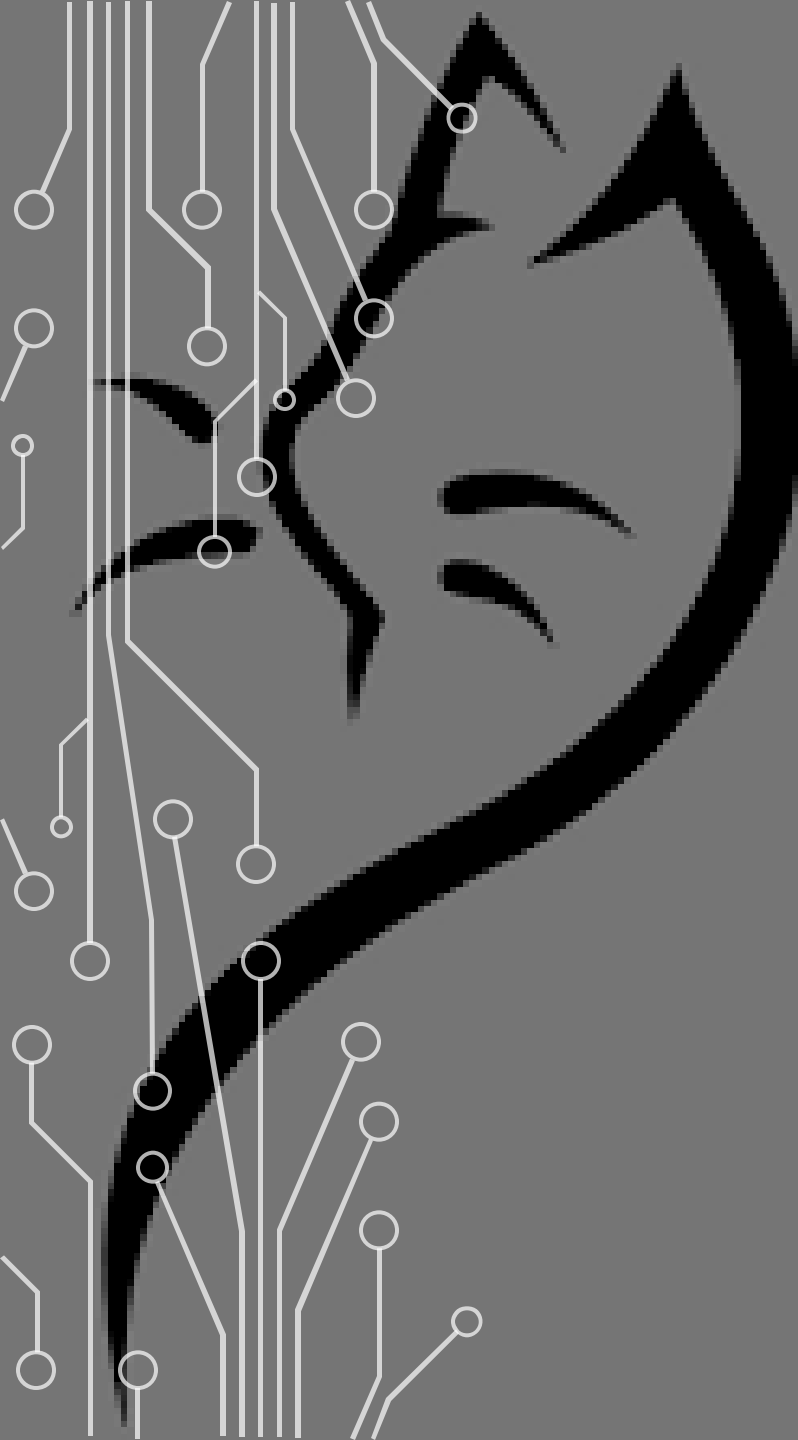
TESTING AND EVALUATION

- Tested clean code
- Tested eval() vulnerability detection
- Verified JSON response formatting
- PyTest backend tests
- GitHub Actions workflow validation

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GITHUB ISSUES / FUTURE IMPROVEMENTS

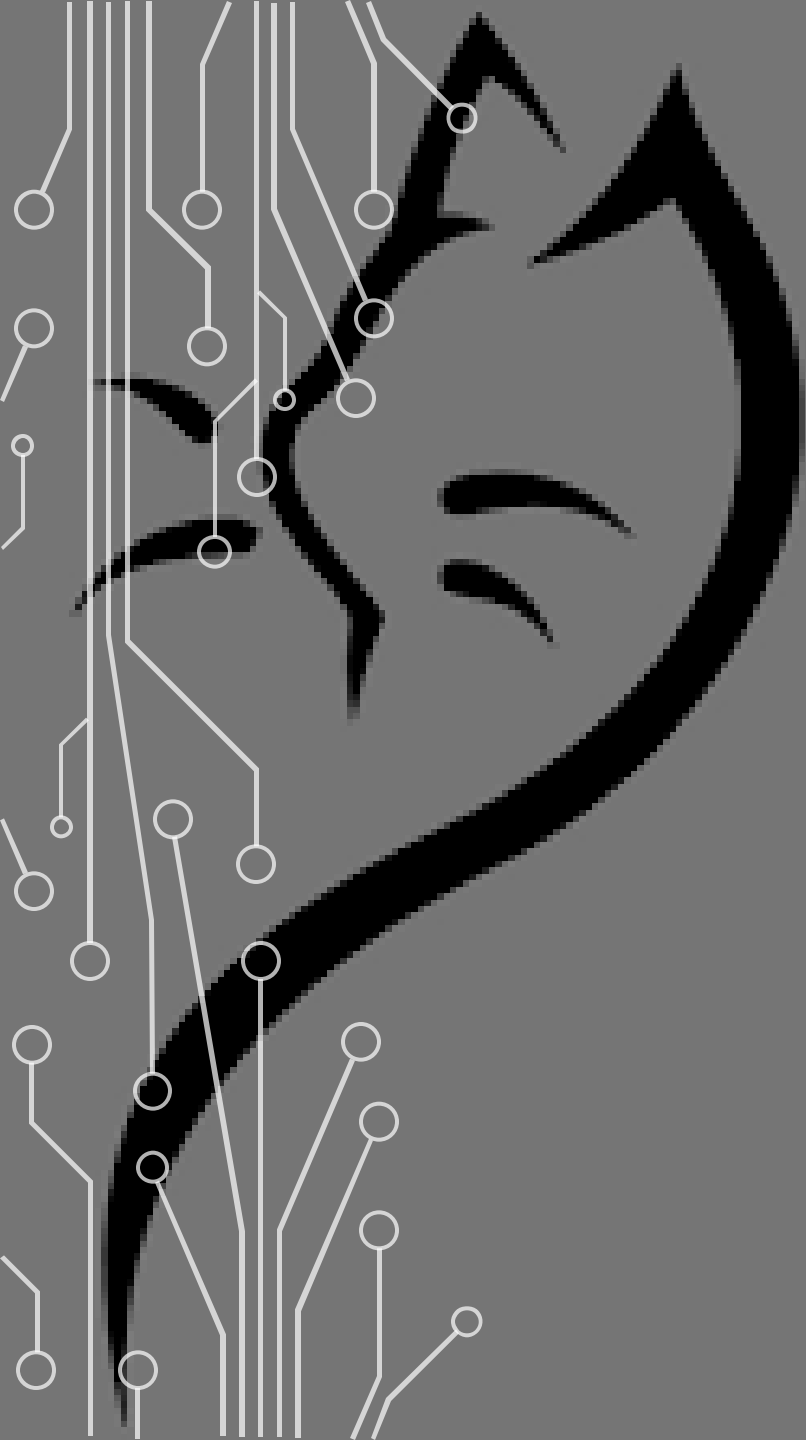
☐	Open	6	Closed	2
☐		Improve handling of malformed AI responses	#8 · amandabradburn-ciss opened now	
☐		Add rate limiting for repeated submissions	#7 · amandabradburn-ciss opened now	
☐		Add support for additional language scanners	#6 · amandabradburn-ciss opened 2 minutes ago	
☐		Add severity color coding to vulnerability results	#5 · amandabradburn-ciss opened 4 minutes ago	
☐		Add backend tests for /submit route	#4 · amandabradburn-ciss opened 2 weeks ago	
☐		Review Bandit findings in backend code	#3 · amandabradburn-ciss opened 2 weeks ago	

GitHub Issues used to document technical debt and future enhancements

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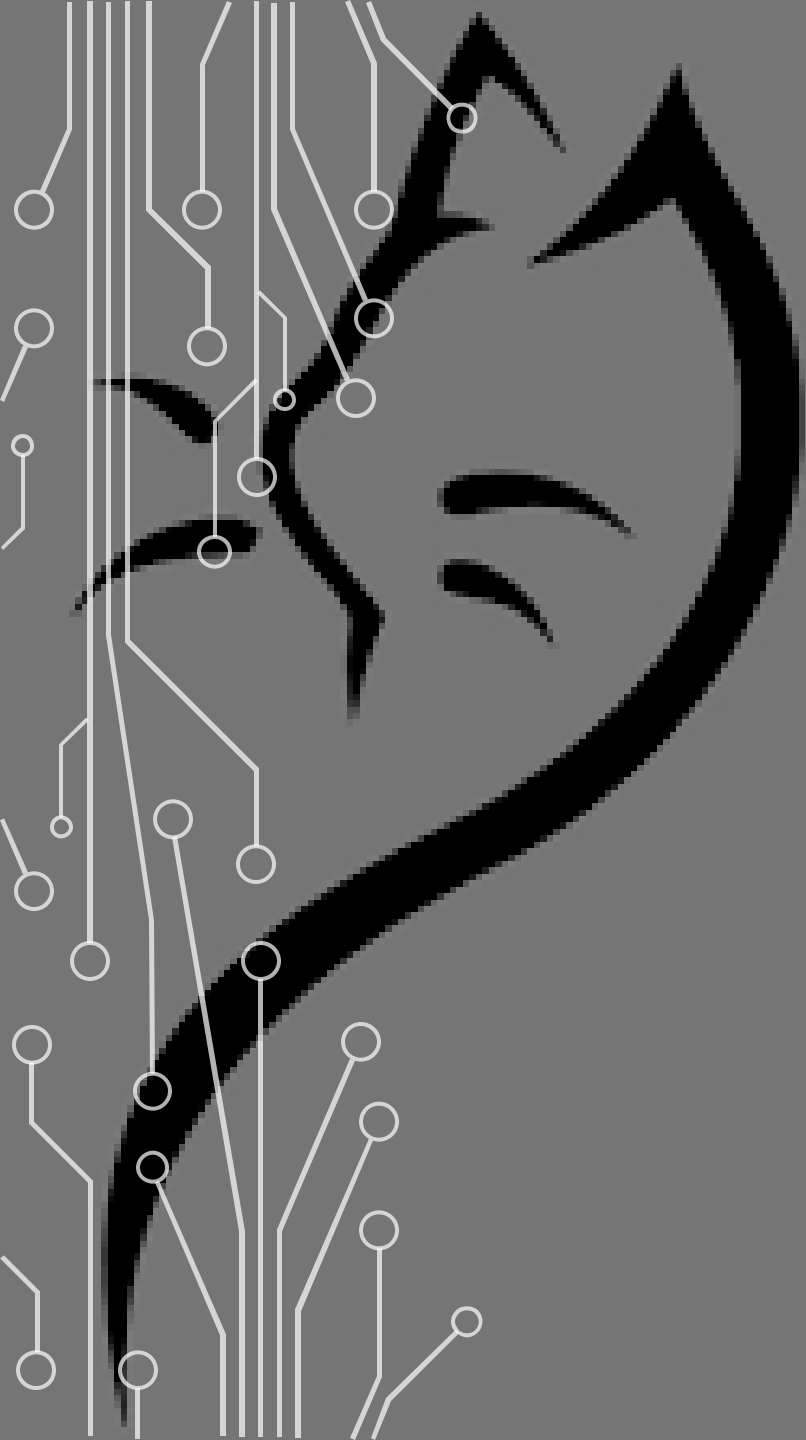
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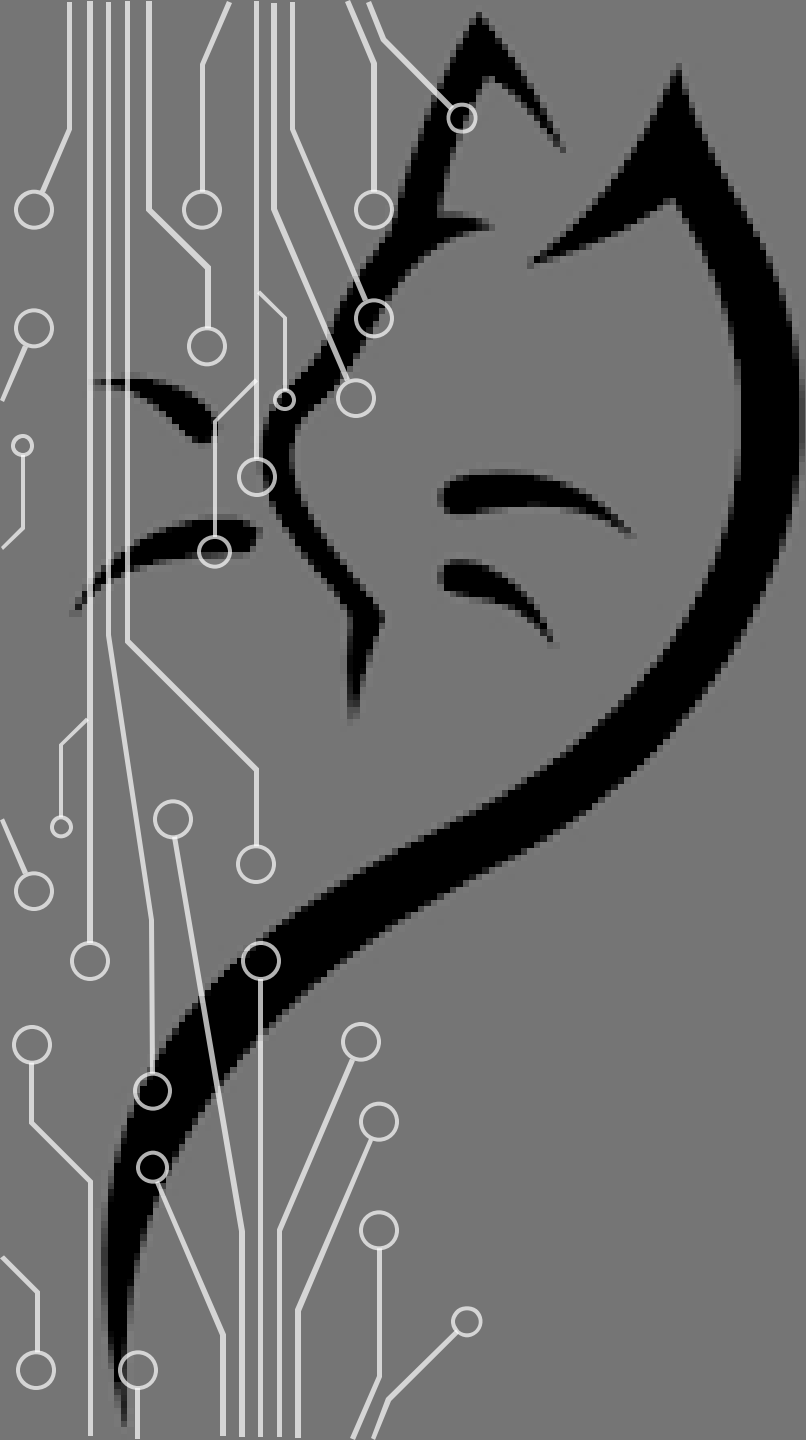
LIVE DEMO

1. Clean code
2. Vulnerable code example
3. AI beta languages
4. CI/CD workflow
5. Error handling demonstration

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FUTURE DEVELOPMENT

- Additional language scanners
- Better frontend visualization
- Expanded monitoring
- Improved AI response validation

COPYCAT — KEY TAKEAWAYS

- CopyCat successfully combines static analysis with AI explanations
- Security-focuses iterative development
- CI/CD and testing improved reliability

Amanda Bradburn — Backend / Security / Integration Developer

Will Davis — AI / LLM Integration Developer

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